

CA-C30L: Mobile Application Development Lab

LAB PROGRAM-2 : SCREEN ORIENTATION

2.Creating an application that displays message based on the screen orientation.

Program description :

The MainActivity.java file contains a button click event that has the code to change the screen orientation .The screen orientation is changed using

setRequestedOrientation(ActivityInfo.SCREEN_ORIENTATION_LANDSCAPE); based on Boolean value isPortrait true or false.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
```

```
xmlns:app="http://schemas.android.com/apk  
/res-auto"
```

```
xmlns:tools="http://schemas.android.com/to  
ols"
```

```
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:orientation="vertical"  
    tools:context=".MainActivity">
```

```
<TextView
```

```
    android:id="@+id/tV1"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Screen Orientation"  
    android:textStyle="bold"/>
```

```
<Button
```

```
    android:id="@+id/btnToggle"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"
```

```
        android:layout_marginTop="60dp"
        android:text="Click to change
Orientation"
        android:onClick="onClick"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.lab2;

import android.content.pm.ActivityInfo;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import
androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends
AppCompatActivity {

    private boolean isPortrait = true; // Track
```

current orientation

```
@Override
protected void onCreate(Bundle
savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
```

```
    Button btnToggle =
    findViewById(R.id.btnToggle);
```

```
    btnToggle.setOnClickListener(new
    View.OnClickListener() {
```

```
        @Override
        public void onClick(View v) {
            if (isPortrait) {
                // Switch to landscape
```

```
                setRequestedOrientation(ActivityInfo.SCREEN
                _ORIENTATION_LANDSCAPE);
```

```
            } else {
                // Switch to portrait
```

```
setRequestedOrientation(ActivityInfo.SCREEN  
_ORIENTATION_PORTRAIT);  
    }  
    isPortrait = !isPortrait; // Toggle state  
    }  
});  
}  
}
```